

Assignment 7 – UVM

This assignment is to exercise how to use the UVM factory. UVM factory can substitute one data type by another data type.

In Assignment 6, you have created one sequence item class, let's assume that you have named it `alsu_seq_item`. This class now has constraint blocks that generate valid and invalid cases. Now you will split them into 2 sequence item classes. The first sequence item class is the same class that you have created `alsu_seq_item` which extends `uvm_sequence_item`, and the second class is `alsu_seq_item_valid_invalid` which will extend `alsu_seq_item`.

Revisit the constraint blocks of the `alsu_seq_item`, if the constraint block generates only valid cases then leave it. If the constraint block generates valid and invalid cases then copy it into `alsu_seq_item_valid_invalid` class, and then adjust this constraint block in `alsu_seq_item` to send only valid cases.

How to replace objected created as `alsu_seq_item` class type in the ALSU environment by `alsu_seq_item_valid_invalid` class type?

- Since both classes are registered in the UVM factory then add the following method call in the test build_phase:
- `set_type_override_by_type(alsu_seq_item::get_type(), alsu_seq_item_valid_invalid::get_type());`

The above call will replace one type with another type whenever any object is created using the `type_id::create` method.

Deliverables:

Run the UVM testbench without the override method call and then rerun after adding the override method. Take screenshots of the waveforms showing that the first run generates valid cases only, then take screenshots of the waveforms that the second run generates valid and invalid cases